

Energy Consumption Assistant

From raw consumption data to AI-powered strategy recommendations



INTRODUCTION

What is this project?

Energy consultants need to quickly understand a customer's electricity consumption - what drives their network cost, where the risk sits, and what to recommend - without being data scientists themselves.

This app aims to help the consultants in dressing up a customers consumption profile, analyze it and extract features that will help the consultant in his task. The app will go further and dress a customer profile and generate a diagnostic and propose strategies to the consultant.



Translate raw meter data into clear behavioral insight

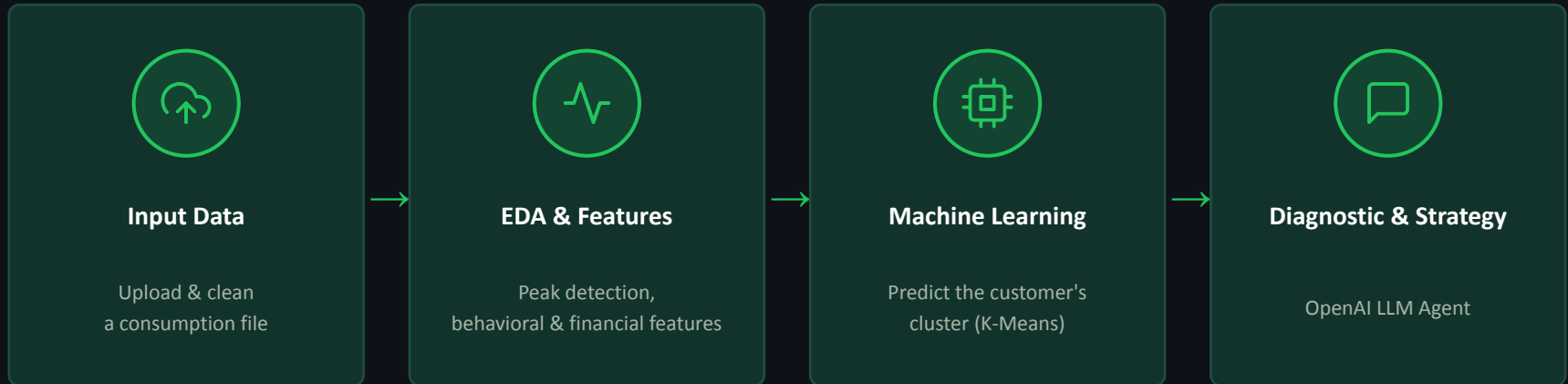


Understand and establish the financial aspect - peaks vs. base



Generate a consultant-ready diagnostic and propose strategies to optimize the consumption

How the App Works - The Pipeline



Raw vs. Generated Data, and Cleaning



Real-data handling

Raw German “Lastgang” meter-file loader (15-minute electricity, hourly gas). Main issue is not enough data to train a ML model with !



Data cleaning

Europe/Berlin timestamps shift between +01:00 and +02:00 across the year (Skip or duplicate records).



Synthetic Customer Generator

Generate random profiles around 5 electricity archetypes (small/medium commercial, industrial, intensive industrial, flexible-peaky), each with coherent gas behavior. Overall I generated 250 customers trying to fill the different archetypes with adapted ratios (more small commercials, less intensive industrial)

Peak Identification & Behavioral Features



Peak identification

Deliberately simple and explainable, not a black-box anomaly detector:

- Baseline = each customer's own median load
- Threshold = each customer's own 95th percentile
- Consecutive readings above threshold = one “peak event”

Every peak can be traced back to a rule a consultant can explain to a customer.



Behavioral features

Discussion with a consultant to determine which features to build:

- Consumption level & variability
- Time-of-day / weekday-weekend shares
- Peak count, ratio, duration, and timing

Still in the process of optimizing to improve the model and define clusters more accurately.

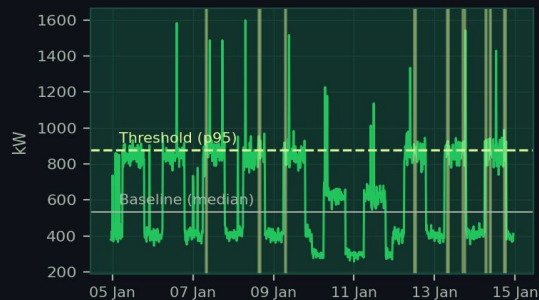
Overview of the pipeline for One Customer

Customer Consumption - Raw input

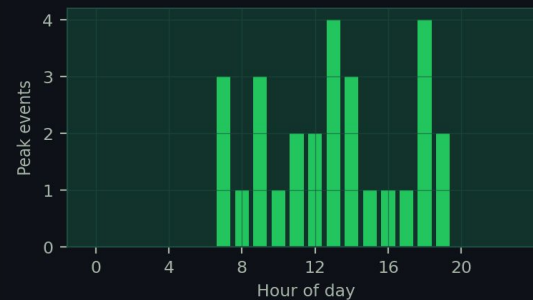


Peak Identification - Feature Engineering

10-day excerpt (5-15 Jan) - zoomed in



Peak Timing - Profiling for ML model



timestamp	power_kw
2026-01-01 00:00	396.5
2026-01-01 00:15	393.6
2026-01-01 00:30	418.5
2026-01-01 00:45	415.5
2026-01-01 01:00	409.9

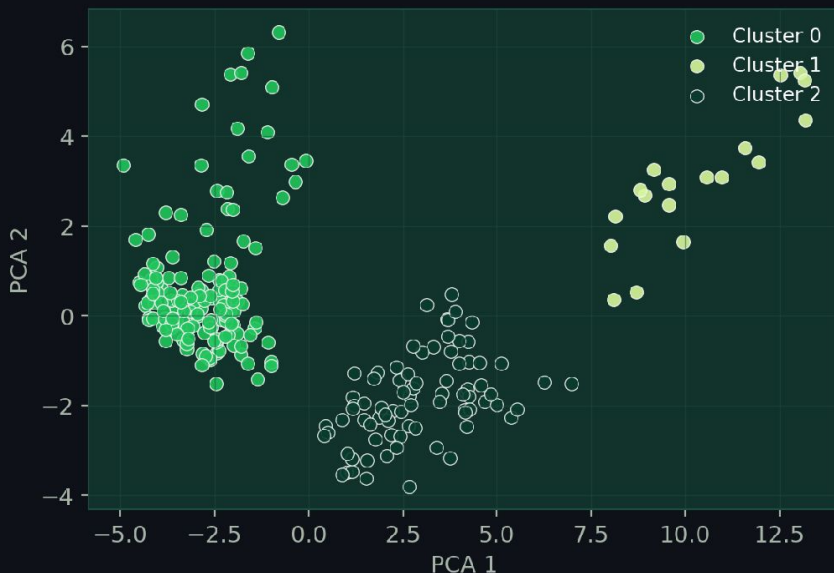
peak_time	value(kW)	ratio
01-07 08:45	941.5	1.76
01-08 16:15	954.7	1.78
01-09 07:30	919.5	1.72
01-12 12:45	910.3	1.70
01-13 09:00	926.7	1.73

feature	value
annual_consumption	4.88 GWh
max_consumption	1,901.9 kW
peak_count	28
morning_peak_share	28.6%
daytime_peak_share	50.0%
weekday_peak_share	100.0%

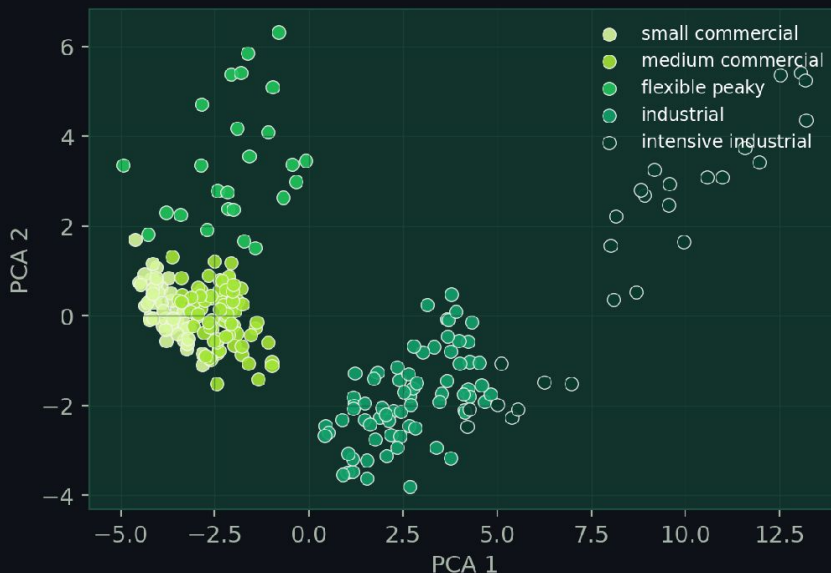
Customer Clustering (K-Means)

Trained blind on 250 generated electricity customers - K-Means never sees the archetype.

**What K-Means Found
(blind, unsupervised)**



**What We Deliberately Generated
(known archetype)**



K-Means cleanly separates industrial-scale from commercial-scale consumption on its own. It doesn't perfectly recover all 5 archetypes ($k=3$), grouping the three lower-consumption profiles together and splitting intensive-industrial across two clusters.

Extracting prices from Official data

What's modeled is the network operator's tariff (Netzentgelte).



Extracted directly from official PDFs

The application extract the price grids directly from pre-downloaded files obtained from the official websites. Westnetz (regional distribution) and Amprion (transmission) publish annual tariffs as PDFs each year. Only Westnetz's distribution-level tariff is applied for now, since every customer here connects at that level.



Two components, selected by utilization hours

Demand charge (Leistungspreis, €/kW) for the customer's annual peak, plus energy charge (Arbeitspreis, ct/kWh) for total consumption. Utilization hours (kWh ÷ peak kW) decide which of 2 tariff brackets applies - this customer: 2,565 h/a → the high-utilization bracket.

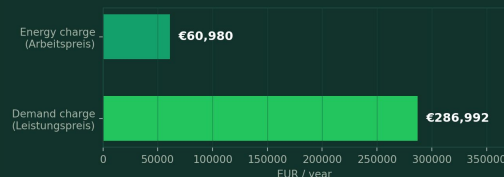
Entgelte für Netznutzung

Jahresleistungspreissystem für Entnahme mit registrierender Lastgangmessung

Netz- oder Umspannebene	Jahresbenutzungsdauer			
	<2.500h/a		≥2.500h/a	
	Leistungspreis €/kW	Arbeitspreis ct/kWh	Leistungspreis €/kW	Arbeitspreis ct/kWh
Höchstspannung mit Umspannung auf Hochspannung	17,18	2,81	76,70	0,43
Hochspannung	13,72	4,57	122,22	0,23
Hochspannung mit Umspannung auf Mittelspannung	14,75	4,85	130,25	0,23
Mittelspannung	16,82	5,71	134,32	1,01
Mittelspannung mit Umspannung auf Niederspannung	17,90	6,57	150,90	1,25
Niederspannung	19,93	8,27	120,43	4,25

Westnetz Preisblatt 1, p.1 - the exact source table for the numbers above

Annual Network Cost Breakdown - this customer



Total
€347,972

AI Diagnostic & Strategy



Structured Output & Guardrails

What it produces

- Profile & behavior summary
- Main cost driver (only when the data supports one)
- Key findings, each tied to a real number
- Consultant talking points
- Caveats: what's missing or uncertain

Guardrails

- Never invents a consumption, price, or savings figure
- Preserves every supplied number exactly
- Distinguishes measured facts from simulated scenarios
- Missing data as a caveat
- Never claims technical feasibility the data doesn't support



Agent Prompt

*The customer features are saved in a JSON file.
Use json.dump to send the customer profile to the OpenAI agent*

```
llm_profile.py

1 def build_user_message(customer_record: dict) -> str:
2     """The user-turn content sent alongside SYSTEM_PROMPT: the record,
3     verbatim, as JSON - nothing added, nothing paraphrased."""
4     return (
5         "Here is the structured record for one customer. Produce the diagnostic "
6         "following the required schema, using only the information below:\n\n"
7         + json.dumps(customer_record, indent=2, default=str)
8     )
9
10
11 # ...called as:
12 messages=[
13     {"role": "system", "content": SYSTEM_PROMPT},
14     {"role": "user", "content": build_user_message(customer_record)},
15 ],
```

Live Demo



What's Next?



Gather more real customer data

Start training the ML model with real case data.



Refine peak identification

Move beyond a fixed 95th-percentile cutoff. Explore more the literature, maybe create another ML model here ?



Deepen feature extraction

Will come with more real data (refine which features to extract to improve the ML model)



2nd LLM 'Chatbox'

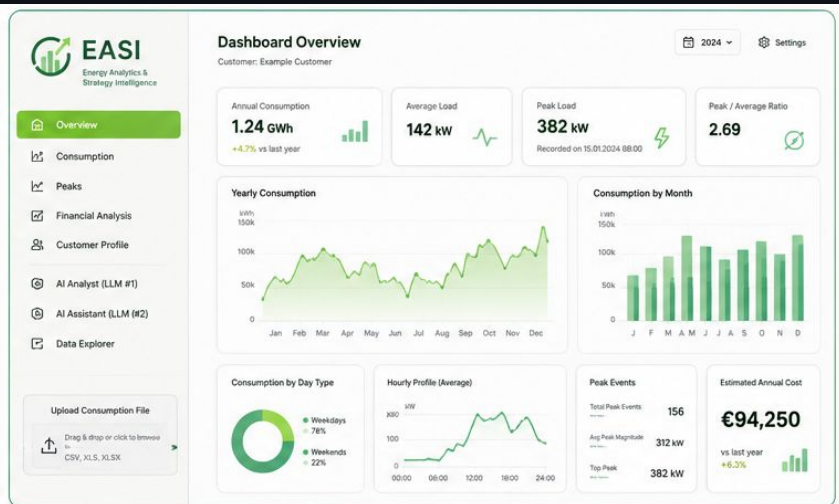
Let a customer ask questions about his consumption: "what if I cut my peak by 15%?" and get a live, calculated answer.



Turn it into a product: EASI

Energy Analytics & Strategy Intelligence (EASI)- take the app from prototype to product.

End Goal : Create EASI



Thank You !

Questions ?

